

Parth Varsani

linkedin.com/in/parthvarsani | +1 (720)-401-8251 | parth.varsani@colorado.edu

Summary

Embedded Firmware Engineer with 2.5+ years of industry experience developing firmware and Linux drivers for ARM-based SoCs. Strong foundation in bare-metal C, RTOS concepts, memory-mapped I/O, DMA, and interrupt handling, with additional experience in char device driver development, BSP customization, and Yocto build systems.

Education

University of Colorado Boulder, CO

Aug 2024 – May 2026

Master of Science in Electrical & Computer Engineering (Embedded Systems)

GPA: 3.5/4.0

Relevant Coursework: Principles of Embedded Software, Wireless LANs, IoT & Embedded Firmware, Advanced Embedded Software Development, Concurrent Programming, Embedded Systems Design, Real-Time Embedded Systems

BVM Engineering college, India

Jul 2018 – Jun 2022

Bachelor of Technology in Electronics Engineering

CGPA: 7.3/10.0

Relevant Coursework: Micro-controllers, Sensors & Transducers, Embedded & IOT System Design, Digital Logic Design

Technical Skills

Programming & OS: C, GNU Make, Shell Scripting, ARM Assembly Language, FreeRTOS, Embedded Linux

Communication Protocols: Wi-Fi (802.11 n/ac/ax), RS232, CAN, I2C, SPI, UART, BLE

Hardware: nRF52840, STM32F4, EFR32 Blue Gecko, AT89c51, Raspberry Pi4B, NXP i.MX EVKs (Cortex-A53), SD8987

Tools and IDEs: Git, iPerf3, QEMU, Perf, Wireshark, LANforge, Simplicity Studio, MCUXpresso, STM32CubeIDE, Networking utilities, Strace, Trace32, GDB, kGDB, GCC, Altium Designer, Oscilloscope, Logic Analyzer, Multimeter, Yocto, Buildroot

Experience

CableLabs Inc. | Wi-Fi Intern | Louisville, CO, USA

May 2025 – Aug 2025

- Tuned EDCA (AIFSN, CWmin/CWmax, TXOP) parameters in the mwifiex driver stack on Linux to optimize MAC-layer contention behavior for vo/vi ACs, reducing inter-transmission time (ITT) by 35% across multi-station 802.11ac testbeds.
- Built a Python PCAP analysis pipeline parsing radiotap and 802.11 MAC headers to extract per-AC ITT, TXOP duration, and retry distributions, enabling closed-loop parameter tuning that previously required manual Wireshark inspection.

Volansys – An ACL Digital Company | Associate Firmware Engineer | Ahmedabad, India

Jan 2022 – Jul 2024

- Root-caused and fixed an endianness bug in the custom mwifiex driver channel-scan path on NXP i.MX 8MQ (SD8987), tracing corrupted channel descriptors through cfg80211 scan results using GDB and kernel ftrace, patched with le16_to_cpu macros in the scan response parser.
- Debugged Wi-Fi connectivity, roaming, and throughput regressions on SD8997 across driver/firmware boundaries, isolating issues to specific layers (mac80211 vs. vendor driver vs. firmware) through log correlation and PCAP analysis.
- Automated Wi-Fi firmware validation by implementing an /etc/init.d startup daemon that executed scripted connect/disconnect test cycles with varied security and channel parameters, improving repeatability and speeding up bring-up testing.

Projects

Real-Time Pan-Tilt Vision Tracking System (Raspberry Pi 4B)

April 2026 - May 2026

- Built a real-time red object tracking system on RPi 4B using a Pi Camera Module (C270) and dual-axis pan-tilt servo platform, implementing HSV-based color segmentation and PID-driven servo actuation to continuously center a detected target in the camera frame.
- Structured the pipeline as four POSIX SCHED_FIFO pthreads forming a multi-rate system, V4L2 capture, HSV segmentation, and PID servo control at 30 Hz, and a health monitor at 1 Hz, achieving stable closed-loop tracking with zero deadline misses across all rate domains.
- Achieved CPU utilization $U = 0.19$, well below the 4-task RM LUB of 0.757, with measured WCET of all 4 services and sub-millisecond inter-service jitter under stressing load & validating hard real-time schedulability with significant margin.

Custom 8051 Development Board & Drivers

Aug 2024 – Dec 2024

- Designed a custom 8051 development board with power, clock, and reset circuitry, connecting all address/data lines via wire-wrap and using a CPLD to differentiate fetches from NVSRAM vs CGRAM.
- Validated system bring-up and peripheral communication by developing assembly and C drivers for UART, I2C EEPROM, SPI, LCD, and ADC, using oscilloscope for current measurement, logic analyzer for SPI/I2C verification, and UART loop back for TX/RX testing.

CAN-Based Proximity Alert System (Raspberry Pi 4B)

April 2025 – May 2025

- Authored a character-device I²C driver in C for the APDS9960 proximity sensor using i2c_rdwr_ioctl(), exposing a periodic-sampling interface through a FIFO IPC to userspace, streaming real-time proximity data on embedded Linux.
- Built a Yocto meta-layer integrating the I²C driver, MCP2515 CAN device-tree overlay, GPIO handling, and systemd services, enabling reproducible multi-node boot and clean deployment from a single bitbake image recipe.
- Designed a two-node distributed application over SocketCAN RAW sockets with TX/RX state machines and threshold-triggered actuation, allowing two Raspberry Pis to exchange sensor events with fault-resistant LED response.